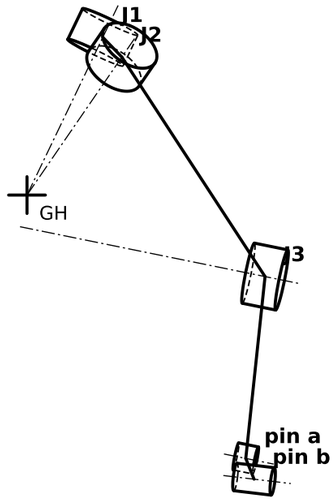
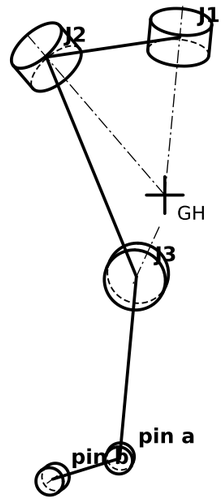
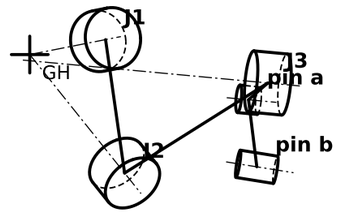
FROM BEHIND (z right, y up)



FROM THE RIGHT SIDE (x right, y up)



FROM ABOVE (z right, x up)



JOINT AXES IN THE BODY FRAME (origin: GH centre; x anterior, y up, z lateral; rest pose; mm)

JOINT	AXIS i, j, k (outward)	ELEV.	AZIM.	REF. POINT x, y, z	FACE FROM GH	AXIS OFF GH	HUB H	∠ TO PREV.	GAP TO PREV.	HUB CTR DIST.
J1	+0.0852 +0.8980 +0.4316	63.90°	78.84°	4.40, 46.43, 22.32	51.70	0.00	11.15			
J2	-0.5776 +0.6727 +0.4624	42.28°	141.32°	-35.41, 41.24, 28.35	61.31	0.00	11.15	41.02°	0.00	44.42
J3	-0.0912 -0.1985 +0.9758	-11.45°	95.34°	-8.75, -25.47, 72.19	76.30	10.72	11.15	68.26°	7.67	90.36
pin a	-0.0912 -0.1985 +0.9758	-11.45°	95.34°	-14.43, -85.21, 68.13	84.72	70.24	6.58	0.00°	59.55	59.87
pin b	-0.1575 -0.1168 +0.9806	-6.71°	99.12°	-35.62, -91.90, 67.96	82.99	86.29	12.40	6.04°	20.38	23.05

REF. POINT: on the axis at the joint's skin-side face (the lower member's face). FACE FROM GH: that face's distance from GH along the axis.
AXIS OFF GH: closest distance from GH to the axis line (0 = the axis passes through the shoulder centre). HUB H: skin-side face to crown.
ELEV.: angle above horizontal. AZIM.: angle in the horizontal plane from anterior (+x) toward lateral (+z).
∠ / GAP TO PREV.: angle and closest distance between this axis and the previous joint's axis (the link between them). HUB CTR DIST.: straight distance between hub mid-points.

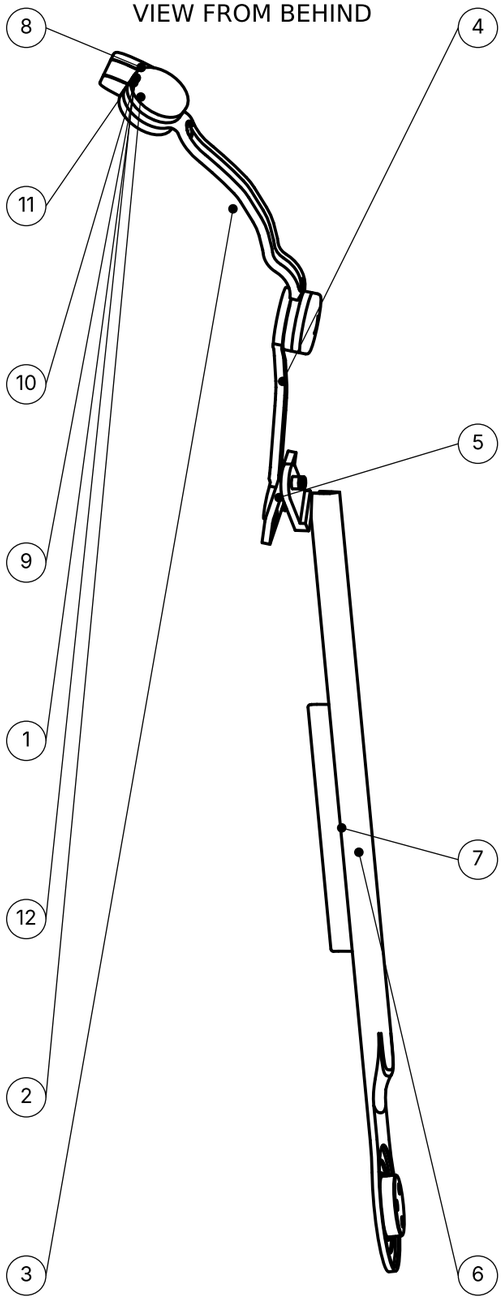
NOTES

1. Use this sheet with the part sheets: each part's END POSITIONS are its slice of this table, in its own frame.
2. J1, J2 axes pass through GH (spherical chain); J3 is offset. Knee pins are parallel to J3.
3. Rest pose of the built CAD (arm at the side). Joint angles change with the arm; the part-frame geometry does not.

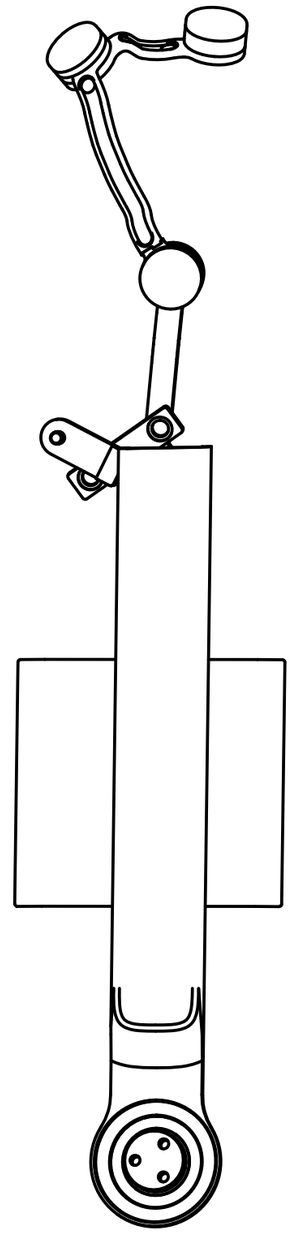
THIRD ANGLE

TITLE **Pin knee H1b, kinematic layout**		DRAWING NO. **EXO-B-KIN**
DESIGN skin-normal-3r-pin-knee H1b	PART NO. **B-KIN**	REV / STATUS A · for review
MATERIAL –	MASS –	QTY PER ASSEMBLY 1
FINISH –	SCALE 1:2	SHEET 1 / 10
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE joint axes from the part STEP files

VIEW FROM BEHIND



VIEW FROM THE RIGHT SIDE



PARTS LIST

ITEM	PART NO.	DESCRIPTION	QTY
1	D8-01	Scapula base (J1 upper member)	1
2	D8-02	Link 1 (J1 to J2)	1
3	D8-03	Link 2 (J2 to J3)	1
4	D8-04	Member 3 (J3 disc + E strap)	1
5	D8-05	Knee link	1
6	D8-06	Arm pad + pad mount + BOTTOM	1
7	D8-07	Upper-arm cuff plate	1
8	H-01	Dome cap Ø20, M18×0.5	3
9	H-02	Hub outer spacer	3
10	H-03	Hub inner spacer, ground (preload)	3
11	H-04	Hub ring nut M7×0.5	3
12	MR117-2RS	Ball bearing 7 × 11 × 3	6
13	GSM-0405-03	iglidur G sleeve 4 × 5.5, faced to 1.89	2
14	GTM-0408-005	iglidur thrust washer 4 × 8 × 0.5	2
15	DIN 6799-3.2	Retaining E-clip (knee pins)	2
16	ISO 10642 M4×6	Countersunk screw (cuff plate to BOTTOM)	6

Materials and finishes: on each part sheet.

NOTES

1. Right shoulder, rest pose. J1, J2, J3 are the three counted joints (Ø20 hubs, sheet HUB). The two knee pins align the cuff and are not counted; the arm pose sets their angles.
2. J1 seats on your scapula handler at the acromion top (bracket on the J1 M18×0.5 spigot).
3. UPGRADE 1 (rule 2026-10-17): knee pins on MR63ZZ pairs instead of iglidur: Ø6 J6 housings, pins ISO 8734 Ø3 m6 pressed in Ø3 H7 seats, bearing seats Ø3 j5, e-clip DIN 6799 RS 2.3 (the lead's knee: bosses Ø9 × 5.5, pin stack 9.75). Keep the pin centres: J3 to pin a 59.55, pin a to pin b 28.24 (sheets D8-04, D8-05).
4. UPGRADE 2: physical knee stop, knee angle 30.46° to 149.54° (used 103.0° to 147.3°): a tooth on the knee link against the E strap.
5. UPGRADE 3: physical J2 stop keeping the J1-J3 axis angle in 32.2° to 104.3° (5° inside the dyad limits 27.24° / 109.28° set by the link angles 41.02° and 68.26°): J2 turned 21.6° to 149.7° from its folded position (J1, J2, J3 axes coplanar, J3 nearest J1), on the branch drawn. At rest the J1-J3 angle is 76.4°. Use a dowel in an arc slot on J2's own angle.
6. Verified score 90.11 (gap 27.20), held-out gap 29.12; complexity 0. Bought parts (items 12-16) are listed, not drawn. Generated from the STEP files of the built CAD; status FOR REVIEW.

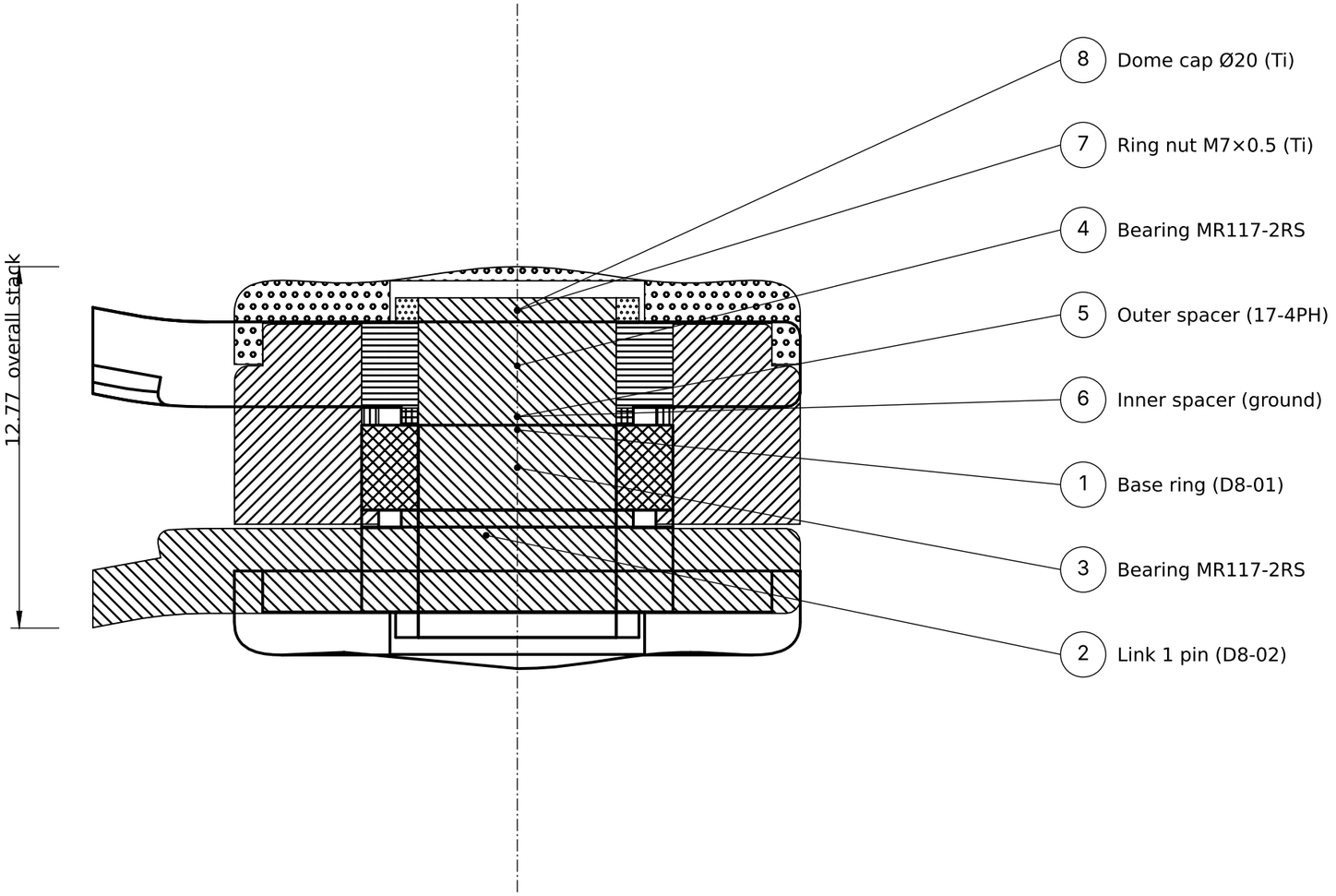
THIRD ANGLE

TITLE: **Pin knee H1b, general arrangement**		DRAWING NO.: **EXO-B-GA**
DESIGN: skin-normal-3r-pin-knee H1b	PART NO.: **B-ASSY**	REV / STATUS: A · for review
MATERIAL: see parts list	MASS: 178.0 g	QTY PER ASSEMBLY: 1
FINISH: see part drawings	SCALE: 1:2	SHEET: 2 / 10
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: pipeline/full-exo/designs/user_pin_knee

NOTES

1. Same stack at J1, J2 and J3: MR117-2RS pair back to back, shaft Ø7 j5 / housing Ø11 J6, preloaded through the ground inner spacer by the M7×0.5 ring nut.
2. 12.25 mm from the skin-side face to the crown (v5 standard, docs/HARDWARE.md).
3. Dome cap threads M18×0.5 onto the ring; at J1 your scapula handler's bracket replaces the cap.
4. Bearings: catalogue MR117-2RS 7 × 11 × 3 (P5 recommended: ring width 0/ −40 µm).
5. Drawn from the J1 hub of the built CAD; section plane through the axis, clipped 15 mm around it.

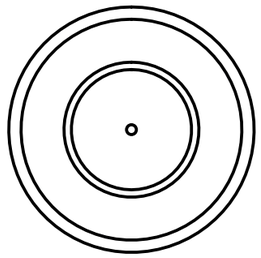
SECTION THROUGH THE HUB AXIS (5:1)



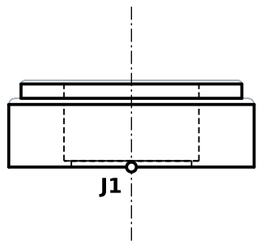
THIRD ANGLE

TITLE **Ø20 hub stack (v5), section**		DRAWING NO. **EXO-B-HUB**
DESIGN skin-normal-3r-pin-knee H1b	PART NO. **HUB-v5**	REV / STATUS A · for review
MATERIAL see part list	MASS see parts list	QTY PER ASSEMBLY 3
FINISH bearing seats unblasted	SCALE 5:1	SHEET 3 / 10
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE pipeline/full-exo/designs/user_pin_knee

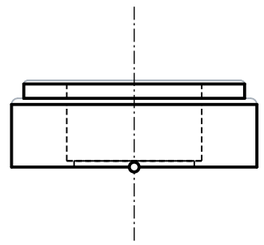
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)

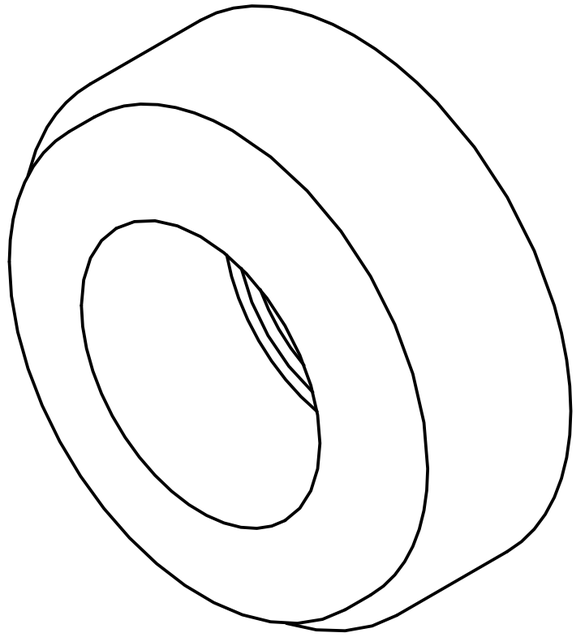


END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J1	0.00	0.00	0.00	0.00°	0.00°	-0.00°	0.00°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

BUILT PART, ISOMETRIC (5:1), for reference



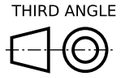
END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J1	Ø9.80	hole	bore	0.00	0.50
J1	Ø20	hub ring OD	pin / boss	0.00	5.10
J1	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
J1	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
J1	faces at s =	0.00, 0.50, 5.60, 7.07			

NOTES

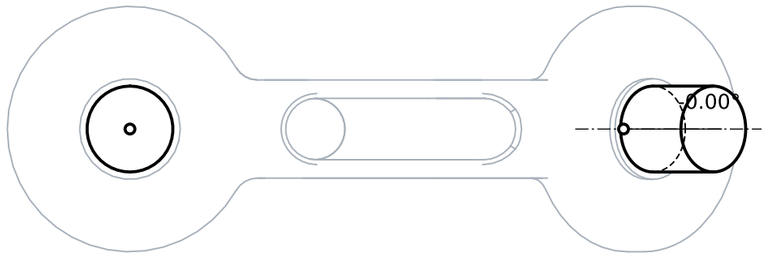
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 29_D8-01_scapula_base_AT_THE_J1_RING.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Your scapula handler's bracket threads onto the M18×0.5 spigot in place of the dome cap; no tab, no ears.
5. 5-axis machined.

THIRD ANGLE

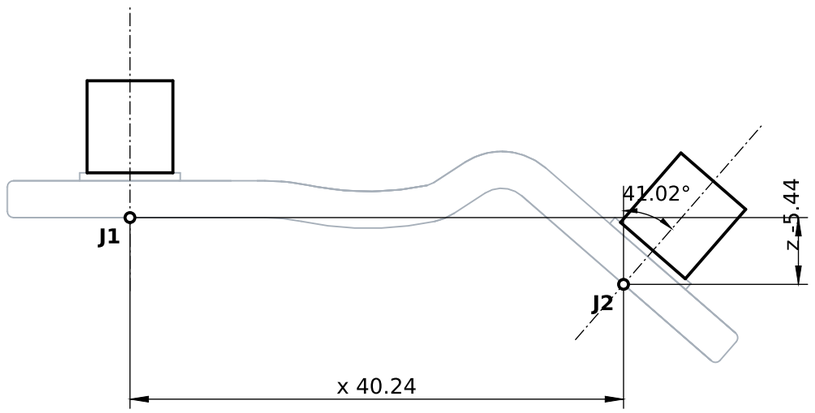


TITLE **Scapula base (J1 upper member)**		DRAWING NO. **EXO-B-D8-01**
DESIGN skin-normal-3r-pin-knee H1b	PART NO. **D8-01**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 6.50 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 4 / 10
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 29_D8-01_scapula_base_AT_THE_J1_RING

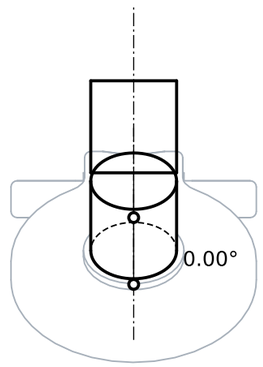
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J1	0.00	0.00	0.00	0.00°	0.00°	-0.00°	0.00°
J2	40.24	0.00	-5.44	41.02°	-0.00°	41.02°	-0.00°

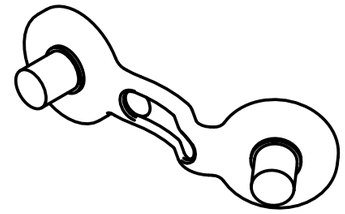
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
J1–J2	41.02°	0.00	-51.70	-61.31

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (1:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J1	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
J1	faces at s =	0.00, 3.00, 3.65, 11.15			
J2	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
J2	faces at s =	0.00, 3.00, 3.65, 11.15			

CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
part	Ø5.00	hole	at 15.1, 0.0, 1.8		len 0.62

NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 18_D8-02_link_1.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. 8 × 3 bar section on a line/arc chain, 5-axis machined from solid (bends tighter than the 15 mm bent-bar rule).
5. Integral Ø7 pins carry the J1 and J2 bearing pairs.

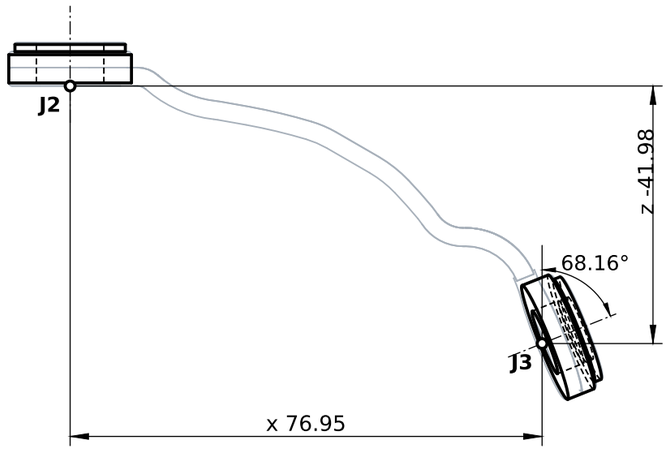
THIRD ANGLE

TITLE **Link 1 (J1 to J2)**		DRAWING NO. **EXO-B-D8-02**
DESIGN skin-normal-3r-pin-knee H1b	PART NO. **D8-02**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 13.45 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 5 / 10
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 18_D8-02_link_1.step

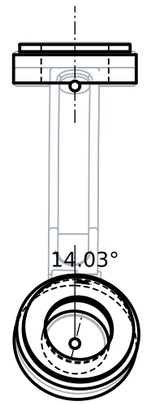
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J2	0.00	0.00	0.00	0.00°	0.00°	-0.00°	-0.00°
J3	76.95	0.00	-41.98	68.26°	5.72°	68.16°	14.03°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
J2–J3	68.26°	7.67	-72.52	-82.43

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (1:2), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J2	Ø9.80	hole	bore	0.00	0.50
J2	Ø20	hub ring OD	pin / boss	0.50	5.10
J2	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
J2	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
J2	faces at s =	0.00, 0.50, 3.00, 5.60, 7.07			
J3	Ø9.80	hole	bore	0.00	0.50
J3	Ø11 J6	MR117-2RS housing	bore	0.50	7.07
J3	Ø20	hub ring OD	pin / boss	0.50	5.10
J3	M18×0.5-6g	thread (cap / handler)	pin / boss	5.60	6.77
J3	faces at s =	0.00, 0.50, 3.00, 5.60, 7.07			

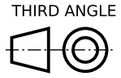
CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
part	Ø5.00	hole	at 15.0, -0.1, 0.4		len 0.67

NOTES

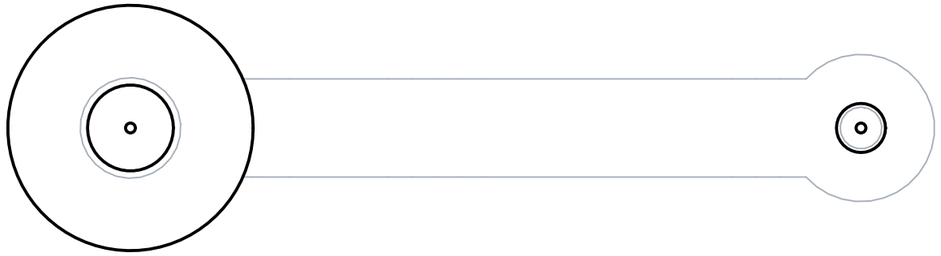
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 19_D8-03_link_2.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. 8 × 3 bar section, 5-axis machined to hug the model strap; flies over the knee and the pad.
5. J2 and J3 rings carry the bearing housings.

THIRD ANGLE

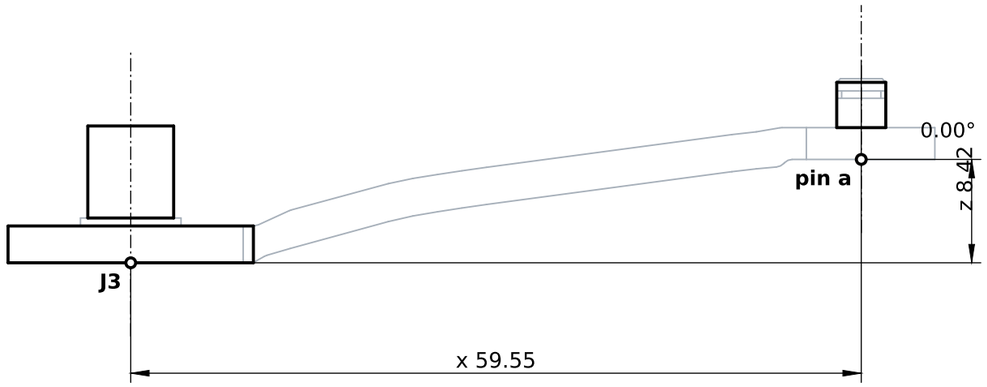


TITLE **Link 2 (J2 to J3)**		DRAWING NO. **EXO-B-D8-03**
DESIGN skin-normal-3r-pin-knee H1b	PART NO. **D8-03**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 19.44 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 1:1	SHEET 6 / 10
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 19_D8-03_link_2.step

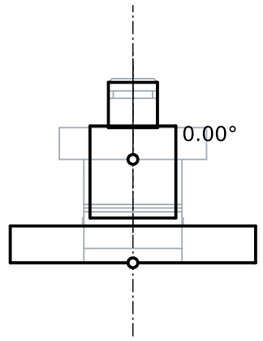
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)



END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
J3	0.00	0.00	0.00	0.00°	0.00°	0.00°	0.00°
pin a	59.55	0.00	8.42	0.00°	0.00°	0.00°	0.00°

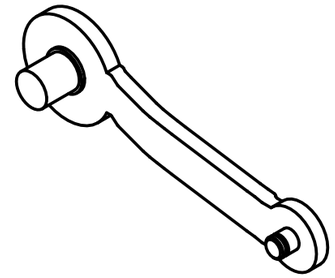
x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

ENDS	ANGLE	AXIS GAP	t1	t2
J3–pin a	parallel	59.55	–	–

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

BUILT PART, ISOMETRIC (1:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
J3	Ø20	hub ring OD	pin / boss	0.00	3.00
J3	Ø7 j5	MR117-2RS seat (integral pin)	pin / boss	3.65	11.15
J3	faces at s =	0.00, 0.02, 3.00, 3.65, 11.15			
pin a	Ø4 h9	pin (integral, hard anodise)	pin / boss	2.59	6.28
pin a	faces at s =	0.00, 2.59, 4.98, 5.58, 6.58			

NOTES

1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 20_D8-04_member_3.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. E strap 8 × 3 from one side-view sketch of lines and tangent arcs; E boss Ø12 carries integral pin a (Ø4 h9).

THIRD ANGLE

TITLE: **Member 3 (J3 disc + E strap)**		DRAWING NO.: **EXO-B-D8-04**
DESIGN: skin-normal-3r-pin-knee H1b	PART NO.: **D8-04**	REV / STATUS: A · for review
MATERIAL: Ti-6Al-4V (Grade 5)	MASS: 11.87 g	QTY PER ASSEMBLY: 1
FINISH: bead blast; mask bores, pins, threads	SCALE: 2:1	SHEET: 7 / 10
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: 20_D8-04_member_3.step

END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
pin a	0.00	0.00	0.00	0.00°	0.00°	-0.00°	-0.00°
pin b	28.24	-0.00	0.00	0.00°	0.00°	-0.00°	-0.00°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

AXIS RELATIONS

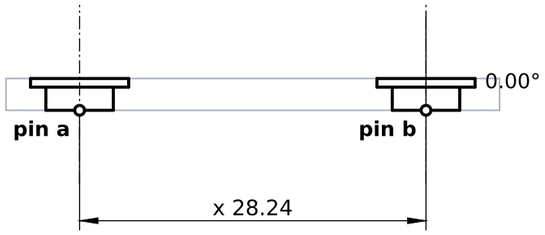
ENDS	ANGLE	AXIS GAP	t1	t2
pin a–pin b	parallel	28.24	–	–

AXIS GAP: closest distance between the two axis lines (0 = they meet).
t1, t2: where that closest point lies along each axis from
the end's reference point (negative = toward the skin).

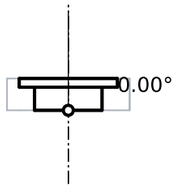
TOP (X right, Y up)



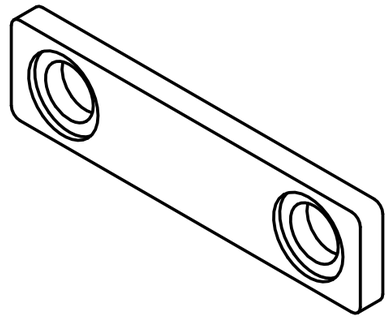
FRONT (X right, Z up)



RIGHT (Y right, Z up)



BUILT PART, ISOMETRIC (2:1), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
pin a	Ø5.5 H7	iglidur GSM-0405-03 press	bore	0.00	1.89
pin a	Ø8	clip counterbore	bore	1.89	2.59
pin a	faces at s =	1.89			
pin b	Ø5.5 H7	iglidur GSM-0405-03 press	bore	0.00	1.89
pin b	Ø8	clip counterbore	bore	1.89	2.59
pin b	faces at s =	1.89			

NOTES

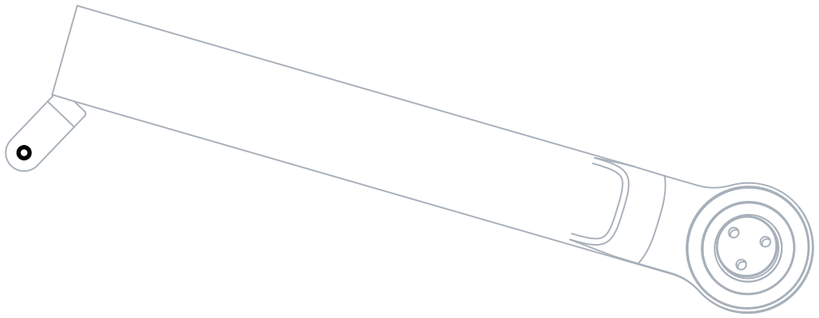
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 24_D8-05_knee_link.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Flat plate from lines + R1 arcs; Ø5.5 bores take the iglidur sleeves; clip counterbores Ø8 × 0.7.
5. Rule since 2026-10-17: rebuild the knee on MR63ZZ pairs (Ø6 J6 housings) with a physical knee stop.

THIRD ANGLE

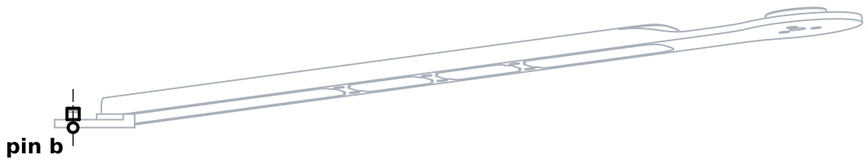


TITLE **Knee link**		DRAWING NO. **EXO-B-D8-05**
DESIGN skin-normal-3r-pin-knee H1b	PART NO. **D8-05**	REV / STATUS A · for review
MATERIAL Ti-6Al-4V (Grade 5)	MASS 3.90 g	QTY PER ASSEMBLY 1
FINISH bead blast; mask bores, pins, threads	SCALE 2:1	SHEET 8 / 10
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 24_D8-05_knee_link.step

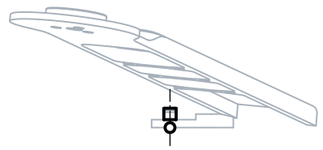
TOP (X right, Y up)



FRONT (X right, Z up)



RIGHT (Y right, Z up)

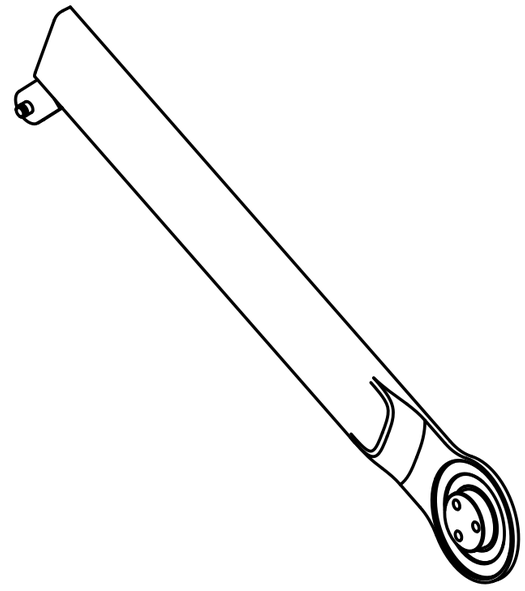


END POSITIONS (part frame)

END	x	y	z	TILT	AZIM.	∠XZ	∠YZ
pin b	0.00	0.00	0.00	0.00°	0.00°	0.00°	0.00°

x, y, z: the end's reference point (on its axis, at its skin-side face).
TILT: true angle between this end's axis and END 1's axis (Z).
AZIM.: direction of that tilt, measured in the XY plane from X toward Y.
∠XZ, ∠YZ: the axis projected in the FRONT and RIGHT views, from Z.
All axes point away from the skin.

BUILT PART, ISOMETRIC (1:2), for reference



END STACKS (along each end's own axis, s from its skin-side face, mm)

END	SIZE / FIT	FUNCTION / FACE LEVELS	TYPE	FROM s	TO s
pin b	Ø4 h9	pin (integral, hard anodise)	pin / boss	2.59	6.28
pin b	faces at s =	0.00, 2.59, 4.98, 5.58, 6.58			

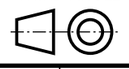
CLOCKED AND OTHER FEATURES (angle about the end's axis from its X reference: toward the other end; END 1: X)

END	SIZE / FIT	FUNCTION	RADIUS	ANGLE	LEVEL s
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 242.0, -29.4, 35.3		len 5.60
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 231.7, -26.4, 33.9		len 5.60
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 148.8, 3.3, 20.9		len 3.50
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 120.2, 11.6, 17.0		len 3.50
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 91.6, 19.8, 13.2		len 3.50
part	Ø5.10	hole	at 236.1, -31.0, 33.3		len 2.50
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 234.1, -36.8, 35.4		len 5.60
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 143.7, -13.4, 22.5		len 3.50
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 115.1, -5.1, 18.6		len 3.50
part	M4 tapped	tap drill Ø3.3, thread M4×0.7	at 86.6, 3.1, 14.8		len 3.50

NOTES

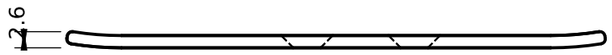
1. Ends are the interfaces to keep: positions, angles and end stacks above. The bar between them is free: built path in STEP 25_D8-06_BOTTOM_+_arm_pad_+_pad_mount,_one_.step.
2. Keep the bar 8 × 3 mm section minimum, 2-14 mm off the skin, clear of the neighbouring parts over the range of motion.
3. Fits: hub Ø7 j5 / Ø11 J6 (MR117-2RS pairs); knee Ø3 j5 / Ø6 J6 (MR63ZZ). General tolerances ISO 2768-mK.
4. Your BOTTOM bar, cut square to the arm 3.7 mm distal of pin b and merged with the pad mount; the only one of your parts that is modified.
5. Pin b is integral (Ø4 h9).

THIRD ANGLE

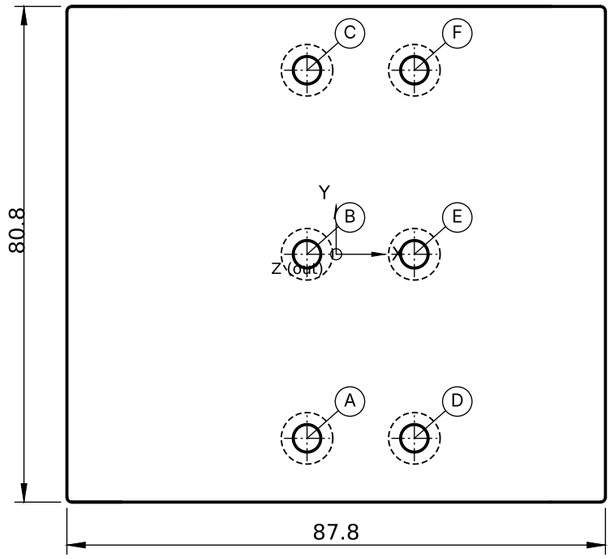


TITLE **Arm pad + pad mount + BOTTOM (one part)**		DRAWING NO. **EXO-B-D8-06**
DESIGN skin-normal-3r-pin-knee H1b	PART NO. **D8-06**	REV / STATUS A · for review
MATERIAL 7075-T6	MASS 66.29 g	QTY PER ASSEMBLY 1
FINISH clear anodise (never black); pin b hard anodise	SCALE 1:2	SHEET 9 / 10
GENERAL TOLERANCES ISO 2768-mK · mm · Ra 3.2	DATE 2026-10-18	SOURCE 25_D8-06_BOTTOM_+_arm_pad_+_pad_m

TOP

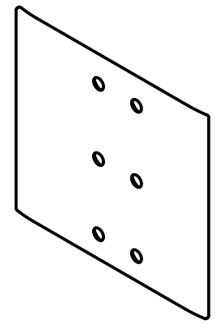


FRONT (looking at the outer face)



RIGHT





ISOMETRIC (1:2)

NOTES

1. Master geometry: STEP file 67_UPPER_ARM_CUFF_(one_sketch.step. Dimensions not shown: take from the model.
2. Bores and pins: the hole table. Fits from the design fit list (hub: Ø7 j5 shaft / Ø11 J6 housing for MR117-2RS pairs).
3. Position of tagged features: Ø0.02 to datum A for hub and pin features, Ø0.05 for others (proposed; review before release).
4. Surface: Ra 0.8 on bearing seats and pins, Ra 1.6 on mating faces, Ra 3.2 elsewhere. Break sharp edges 0.2 max.
5. One sketch (flat land + inner / outer arcs, both side arcs trimmed clear of the trunk), extruded.
6. Fixed to BOTTOM with 6 × M4×6 ISO 10642.

HOLE AND PIN TABLE

Origin: tag A axis, mid-length (datum A). X, Y, Z: part axes as drawn. Centres in mm.

TAG	SIZE / FIT	FUNCTION	TYPE	LEN	X	Y	Z	AXIS i, j, k
A	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	-4.75	-30.00	1.29	+0.000 +0.000 +1.000
B	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	-4.75	-0.00	1.29	+0.000 +0.000 +1.000
C	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	-4.75	30.00	1.29	+0.000 +0.000 +1.000
D	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	12.75	-30.00	1.29	+0.000 +0.000 +1.000
E	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	12.75	0.00	1.29	+0.000 +0.000 +1.000
F	Ø4.5 + CSK 90°	M4 countersunk clearance (ISO 10642)	thru	0.05	12.75	30.00	1.29	+0.000 +0.000 +1.000

THIRD ANGLE

TITLE: **Upper-arm cuff plate**		DRAWING NO.: **EXO-B-D8-07**
DESIGN: skin-normal-3r-pin-knee H1b	PART NO.: **D8-07**	REV / STATUS: A · for review
MATERIAL: AlSi10Mg (DMLS)	MASS: 36.57 g	QTY PER ASSEMBLY: 1
FINISH: bead blast	SCALE: 1:1	SHEET: 10 / 10
GENERAL TOLERANCES: ISO 2768-mK · mm · Ra 3.2	DATE: 2026-10-18	SOURCE: 67_UPPER_ARM_CUFF_(one_sketch.step